

Secure Authentication using Zero Knowledge Proof

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Abstract—Zero-Knowledge Proof is a cryptographic protocol exercised to render privacy and data security by securing the identity of users and using services anonymously. It finds numerous applications; authentication is one of them. A Zero-Knowledge Proof-based authentication system is discussed in this paper. Advanced Encryption Standard (AES) and Secure Remote Password (SRP) protocol have been used to design and build the ZKP based authentication system. SRP is a broadly used Password Authenticated Key Exchange (PAKE) protocol. The proposed method overcomes several drawbacks of traditional and commonly used authentication systems such as a simple username and plaintext password-based system, multi-factor authentication system and others.

Index Terms—Authentication, Zero Knowledge Proof (ZKP), AES, SRP, cryptography

I. INTRODUCTION

Zero-knowledge proofs are an elegant and effective way of providing authentication services and powerful cryptography tools. Zero-Knowledge proof allows limiting the amount of information transferred between two parties in a cryptographic protocol. The idea is to replace "knowledge" with "knowledge about learning." In a system using ZKP, a prover demonstrates to the verifier that some charge (statement) is valid while dripping nothing but the assertion's legitimacy. These proof systems form the backbone of various cryptographic techniques. "Zero-knowledge proof of knowledge." can be explained as - The prover has the job of proving that he/she associates confidence without exhibiting it to the verifier. Zero-knowledge proof is practical where authentication systems need to ensure that there is no disclosure of the secret information to the following party. Zero-knowledge proof has applications in authentication, messaging, storage control, data sharing, file system control and secures sensitive information.

II. ZKP OVERVIEW

Zero-Knowledge Proof (ZKP) is the most cubist and exciting idea in today's applied cryptography. In the 1980s, Charles Rackoff, along with Silvio Micali and Shafi Goldwasser, first suggested 'Zero-Knowledge Proof.' These men have proved that it is achievable to verify that some theories are valid without presenting the sparsest implication of why they are accurate. For instance, a party proving the statement is actual without sharing any hints or information with the second party. ZKP demonstrated to be very valuable in the theory of complexity as well as cryptography. The research area of zero-knowledge proofs for authentication is less explored. Therefore, this particular field has wide scope for research.[1]

Without proffering the authentic meaning of the data, verifying facts from one party by the other party is the notion behind the Zero-Knowledge proof. Such a system is used to validate externally, the verifier not knowing the essential knowledge. The zero-knowledge proof method is verifying data without exchanging the secret keys (passwords) and hence, they cannot be duplicated. This can be explained with an example containing two entities where one has to verify the other entity. In this instance, the target is to find Robert (object) in a famous book series. The catch is to locate Robert in an ocean of things that resemble him. Victor has found Robert but has to prove it to Peggy without letting Peggy know the location of Waldo. Victor cuts a hole in the shape of Robert in an extensive, perplexing film that is opaque and places this film cut-out above the actual display. This explication shows only Robert and none of the surroundings. Robert's coordinates are still unknown relative to the rest of the set. Later, Victor can reflect on the scene below to demonstrate that he used the

primary puzzle. This way, Peggy does not get the location of Waldo but knows that Alice has found him [2].

A. Properties of ZKP

[3] Zero-Knowledge Proof systems are made up of three essential properties: Soundness, Completeness and Zero-Knowledge. Concerning the given example, for better understanding of the following:

a. Soundness — the state of being robust: We can assume Victor is unaware of Robert's exact location and gives arbitrary pieces of information concerning the scene to her proof systems. Then, the holes in her cardboard will exhibit random images bereft of Robert. In simple words, the proof systems of Victor are trustworthy and do not let her fool them.

b. Completeness — the condition of having all necessary parts: So long as Victor can find Robert, he can congruously use his proofs to exhibit Robert in each game. In simple words, the proof systems of Victor can persuade Peggy that he was able to find Robert.

c. Zero-Knowledge — proving without revealing: As Victor demonstrates to Peggy that he has discovered Robert, the only fact shown to Peggy is that "Victor was able to find Robert." The location of Robert is never let out. In simple words, the proof systems of Robert have convinced Peggy about his victory without giving out his knowledge.

B. Real world applications of ZKP

1) Authentication Scheme:

This is the most important application of ZKP. Here, ZKP is used in authentication schemes where a user wants to prove its identity to another user or another party, by having the same confidential data for example, passwords. The user does not wish to disclose confidential data to the other party. After all, here any random numbers are used as passwords. A proof of password with zero knowledge is a unique type of credential of knowledge which gives a shortened size of passwords. The authentication method with ZKP is a more efficient substitute to the classical authentication method. Reduced time and space complexities create it especially unique for usage.

2) Blockchain Technology:

ZKP can be used in blockchain technology, where traditional blockchain systems can be altered with usage of ZKP in order to send unique information preserving the data privacy. ZKP is a robust cryptographic technique used to validate the statements, in spite of users being unknown. The blockchain includes a decentralized cluster of miners running a protocol which is a secured consensus protocol. Thus, let us assume the blockchain is a technology which is trusted for accuracy along with accessibility, except for privacy. Besides, in order to sustain the global journal which keeps the stability, and also executes programs defined by the user.

3) Voting System:

A voting system is used by voters to cast their votes

over the internet. There exist several security flaws in the internet voting system in these recent years. This system should be able to meet the security requirements i.e Confidentiality(state of being secret), Integrity(protection from unauthorized changes), Availability(probability that the system is operative), Anonymity(degree of identity concealment), Reliability(correctness of the system) etc. To meet these security requirements is really a big hurdle. An internet voting system is implemented whose ultimate goal is the authentication of the voter, maintaining the confidentiality and integrity of the ballot.

C. The features of ZKP in aspects of security

ZKP is a cryptographic protocol for two party communication, a prover and a verifier. The prover has to prove the knowledge of the secret which it possesses without disclosing any information of the secret during the process of proving its secret. The secret consists of the secret information such as the credentials. The ZKP has following secure features:

- No knowledge is transferred. The secret cannot be read by the verifier.
- The verifier cannot be tricked by the prover.
- The verifier, for any third party, cannot act as the prover.

III. NORMAL AUTHENTICATION IN WEB APPLICATIONS

For providing secure transmission as well as access to sensitive information, the process of authentication is a necessity. As per the classical approach of the authentication process, the login credentials which are sensitive and critical data are received by the user most of the time, further these credentials are transmitted over HTTPS to the server side from the client side. Usually, these credentials are not stored in the plaintext format, but during the authentication step those credentials are transmitted in plaintext format. In spite of this issue, it is generally neglected by many of the developers. The more often used protocol by developers is HTTPS, by usage of this protocol a secured connection is established where the credentials are transmitted. With the usage of public keys, this security flaw is partially fixed. The traditional way of authentication is, the user is requested to provide its login credentials. Once the user provides the login credentials, those are transmitted straight to the server. When a secured connection is to be established, HTTPS protocol is used. Yet, once the passwords are transmitted, they can be read by the servers, even after they are in hash format and also after adding salt to it.

The commonly found normal authentication system in web applications is by using a simple form for submitting the credentials. Generally, these systems have SSL communication. In systems with security implemented, the hash of the password is generated by hashing the plaintext of password with md5 hash and then it is sent to the server.

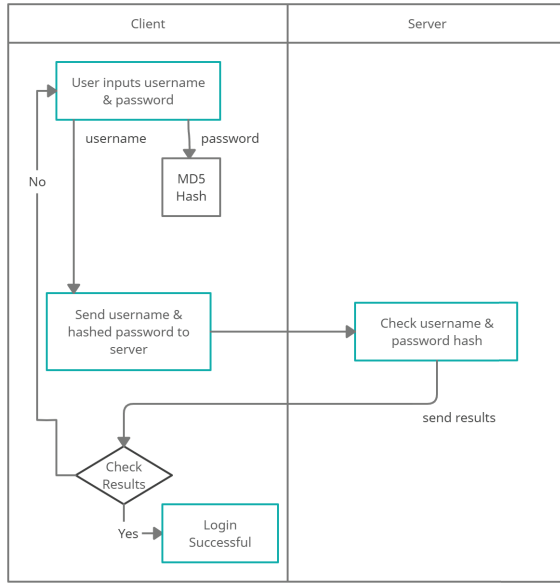


Fig. 1. Normal authentication method workflow

IV. PROBLEM STATEMENT

To provide a secured authentication scheme using Zero Knowledge Proof. Enhance the data security in web applications by using the zero-knowledge proofs. It enables a secured authentication process, without the need of password being transmitted over the channel. The storage of password or the password hashes on the database is prevented by our proposed system. Therefore, even if an attacker gains unauthorized access to the database, the attacker won't gain the password. The algorithm used behind our proposed system is on the concepts of zero knowledge proof. We have used the Advanced Encryption Standard (AES) cryptographic encryption algorithm for the purpose of encryption.

A. Problems with existing system

There exists different vulnerabilities in web applications. This involves web application vulnerabilities, as well as general attack vectors, having these makes the data of the user easily available, for impersonation attacks. Authentication systems are required at every step in this digital world to access online products and services. Loopholes in authentication systems can lead to a number of security and privacy issues. The risks associated with faulty authentication systems can be devastating if not handled properly. Authentication schemes used by various websites fall within a small range of available ways. Problems with some of the commonly used authentication schemes are listed below.

- *Simple password based authentication* : Traditional username and password authentication scheme though easy to implement but scores low on security aspect. It's hard to create a secure password and also it's hard to remember and use those passwords at the right place.

- *Cookie based authentication* : Each time a request is sent on each page from user to the server, the server will need to read data from the database for user validation. It creates a lot of turnover. Also, many modern applications are built not only on the web, but are also native. Implementing cookies for mobile applications is a challenging task. Also it is unreliable because they need to operate in various native environments. Also CSRF attacks are possible on your system when using this scheme of authentication. But this can be overcome if synchronized tokens are used.
- *Passwordless authentication* : This method of authentication is like the two factor authentication scheme without the first step. They use what is commonly called "magic links". Although it is user friendly, if access to your email or mobile is lost through which you are supposed to get the authentication access, which can happen easily, the whole authentication scheme gets compromised.

B. Problems with sending hashed passwords

The problem of sending over hashes of password comes as an attacker just needs to interrupt the communication and gain the information for impersonating as either of the users.[3] Let us take an example, where in a normal authentication method, the login credentials i.e username and the password hash is sent by the user, this enables the hacker to authenticate as a legitimate user, by sniffing the credentials. A more important problem with sending a hash of passwords is the threat of gaining the plaintext of the password using various cracking passwords. If an authentication scheme requires sending a password from the user's system to another, there is scope for the hacker to access the password while it is moving from one system to another or at the server side. Fig 2. shows the problems with sending hashes.

C. Objectives

In this section, a secure approach is proposed for authentication in web applications with the following objectives:

- Provide security to data security using zero-knowledge proof and integrating ECC algorithm.
- To enable the authentication of an user in a secured and efficient manner, without sending the hashed password or the password itself.
- To prevent storage of the hash of password into the database.
- To provide anonymity and a higher degree of privacy to the user.

V. TECHNIQUES AND ALGORITHMS

A. SRP

SRP (Secure Remote Password) is based on key exchange between two parties. A large private key is generated in between two parties in the same way as that of Diffie-Hellman

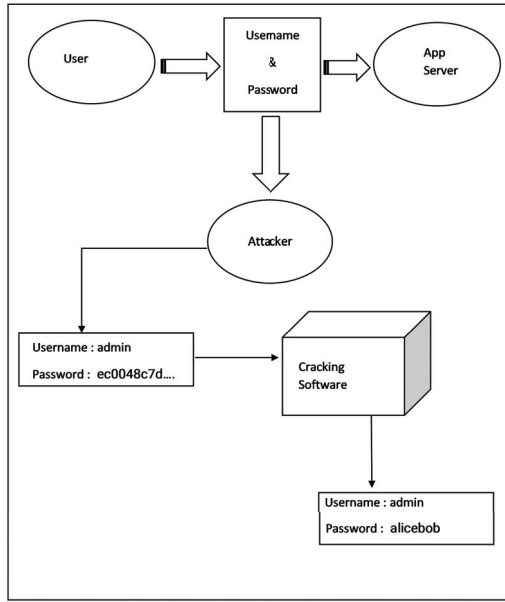


Fig. 2. Problems with sending passwords hashes

key exchange algorithm.[4] The client side has the user login credentials and the server side has the verifier which is generated from the password of the user. Two random numbers are used to generate a shared public key where one number is generated by the client and the other number is unique for each login attempt by the server. The parameters that are used in SRP key computations are as follows:

- Z - Large prime number
- G - generator modulo Z
- salt - salt of user
- Hash() - One way hash function
- usr - Username
- pwd - Password in cleartext
- x - private key generated by using password and salt.
- w - verifier, calculated as Gx .
- i - random public parameter
- p, q - unique private keys generated every time
- Pub_p, Pub_q - Public keys generated
- S_k - Session key
- m, n - Two strings that are combined
- l - Hash(N, g)

The algorithm has three stages, password establishment, key calculations & authentication.

- 1) Password Establishment: The server first calculates the private key, x , generated using the salt and the password entered by the user. For the username, usr , salt, salt, & the verifier, w , are stored into the database.
- 2) Key Calculations: The session key is calculated as follows:

a) $Pub_p = G^p$ where a is randomly generated.

User $\rightarrow$ host - usr, Pub_p

b) $Pub_q = lw + G^q$ Host $\rightarrow$ User i, Pub_q

c) User, Host: $i = \text{Hash}(Pub_p, Pub_q)$

d) User:

$$S_{user} = (Pub_q - l(G^x))^{(q+ix)}$$

e) User:

$$S_{k(user)} = \text{Hash}(S_{user})$$

f) Host:

$$S_{host} = (Pub_p^i)^q$$

- 3) Authentication: User sends Host, proof i.e $h1 = \text{Hash}(Pub_p, Pub_q, S_k)$ to prove its session key. The proof is computed by the host and cross checks it with the User data received. Now, proof $h2, h2 = \text{Hash}(p, h1, S_k)$ is sent by the host to the user as a proof. This is done to enable the host to possess the valid session key. $h2$ is also verified by the User, and verified with that of the Host.

B. AES

AES is a cryptographic algorithm which is widely used for the encryption as well as decryption purposes of the data. AES is approved by the Federal Information Processing Standard. AES is a symmetric block cipher which supports 128-bit data blocks.[5] The key sizes of AES are variable i.e 128, 192 and 256 bits respectively. *State* is a two dimensional array of 16 bytes i.e arranged in 4×4 two dimensional array, where the input data is stored. A round function is used for both the encryption and decryption process that has various asymmetries, and has four various transformations that are byte-oriented. The key's length decides the rounds to be performed. *SubBytes* is a step where the bytes are substituted separately in a non-linear manner by using the substitution table. Substitution table which is used in this step can be inverted. During the decryption process, the inverse of *SubBytes* is *InvSubBytes*. The operation where a left cyclic shift is carried out on the three last rows of *State*, is known as *ShiftRows*. During the decryption process, right cyclic shift is carried out on the same rows mentioned previously. This is known as *InvShiftRows*. The permutation operation which is carried out on all the columns of the *State*, is known as *MixColumns*. The inverse of *MixColumns*, during the decryption is *InvMixColumns*. The only step that operates straight on the round key of AES is *AddRoundKeys*. It is the step where the XOR addition takes place between the round key and the *State*. The Fig 3 represents the process of AES encryption and decryption with the round functions used.

VI. SYSTEM OVERVIEW

In the system that we propose, there are 2 main phases, they are as follows:

- 1) Registration
- 2) Authentication

Registration:

In the Registration phase, the user enters the login details

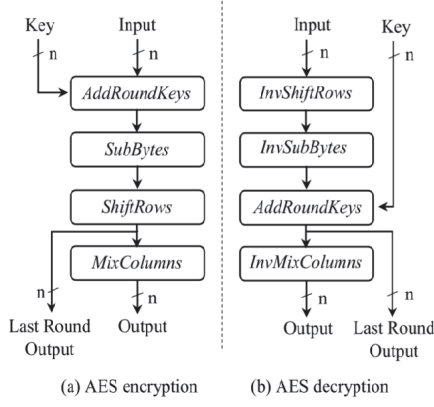


Fig. 3. AES encryption & decryption round functions

i.e the username and the password, on the *register.html* page. On the basis of the user's username and password, the salt, verifier, prime number and the generator are generated respectively. The password is hashed & the prime number and generator are encrypted using the AES algorithm. A key is generated on the basis of the hash of the password. The nonce alongwith tag are generated by using AES, and further stored into the database.

Authentication:

The login page i.e *login.html*, is requested by the user. The user will now enter the username and the password, which he/she has registered previously. Based on the credentials entered by the user, the server will now fetch the salt, verifier, prime number, generator, nonce and tag respectively. The prime number and generator which were encrypted before in the Registration phase, are now decrypted using the key which was derived using the hash of the password. A server session along with its public key is created. The public key of the server session is passed on to the user. Here, a client session is created and processed using the public key of the server session. At the server side, the client session public key is processed with the salt. Now, we verify the validity of the client session and server session. If verified, the user gets authenticated and the *profile.html* page is displayed on the user end.

VII. PROPOSED SYSTEM FOR SECURED AUTHENTICATION METHOD USING ZKP

In this section, the idea of our proposed system will be described briefly. In order to describe and analyze the proposed system in a better way, we are going to use the notations mentioned in the Table 1, below.

Algorithm(1) explains the secure authentication process used in our proposed system.

A. Algorithm(1): Proposed System Protocol

1) Registration:

- a) User enters username, U & password, P .

Notations	Description
U	Username of user
P	Password of user
s	Salt of user
v	Verifier
y	Prime number
g	Generator which is public in nature
$H(P)$	The hash of the password of user
n	Nonce
t	tag
$E(y,g)$	Encryption of y,g by AES & $H(P)$
$D(E)$	Decryption of $E(y,g)$
k	key of $H(P)$
K_s	Server session
A	Public key of server session
K_c	Client session
B	Public key of client session

TABLE I: Notations adopted in the system proposed.

- b) Verifier (v), salt (s) & prime number (y) are generated using P .
- c) The password is hashed using a hash function, H and $H(P)$ is calculated.
- d) $E(y,g)$ is computed using AES & k is derived.
- e) Nonce(n) alongwith tag(t) are generated & stored into the database.

2) Authentication:

- a) User gives an input of U & P .
- b) $D(E)$, $E(y,g)$ is decrypted using k .
- c) Server session, K_s with A , is generated on U , s , v , y .
- d) The server sends A to the user. User generates a client session, K_c using A and generates B .
- e) The user sends B to the server, where B is processed with s .
- f) The server now verifies,
If K_c is true, then verifies K_s
If K_c is false, then terminates, authentication fails
- g) The server next verifies,
If K_s is true, if not then terminate, authentication fails
- h) If both, K_s & K_c are true, then authentication succeeds & user gets authenticated.

Our proposed system is mainly divided into 2 phases, first is the Registration Phase and the second is the Authentication Phase. In our proposed system, we introduce a two party communication between the user which acts as the client and the server.

B. Registration Phase

Here, when the user registers itself, the user enters the username and the password. Based on the credentials entered by the user, the server initializes a set of random numbers : salt, verifier, prime number and the generator. These generated

numbers are complex in nature. The registration phase itself gets subdivided into two steps. First step is the initialization of the salt, verifier, prime number and the generator. Second, is the registration of the user. Server uses a hash function for generating the hash of the password and AES is used to encrypt the prime number and generator. Using AES, we get a key based on the hash of the password. Therefore, additional data named, nonce & tag are generated and stored into the database by the server. Fig 4. denotes the general working flow of the registration phase.

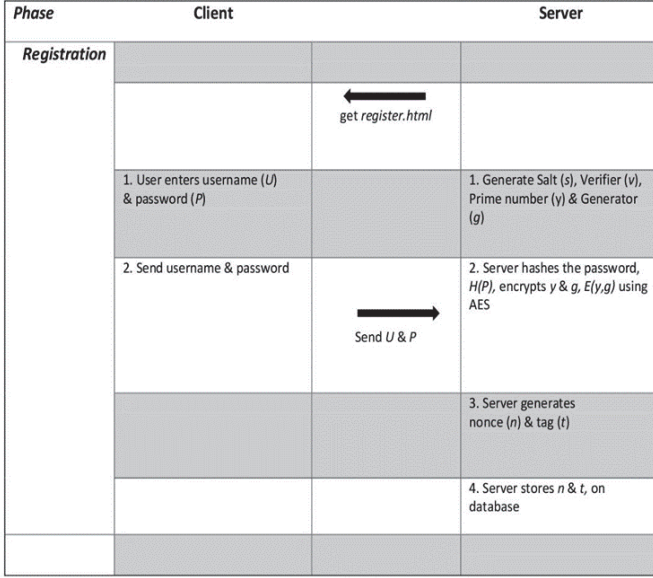


Fig. 4. General structure of the Registration Phase.

C. Authentication Phase

All the users have their accounts registered in the previous phase. In our proposed system, the Authentication phase is split up into two main steps. First is where the session of both user (client) and server are generated with their respective public keys. Second, the validity of the sessions is verified for the authentication to take place. For every authentication attempt, a new session key is generated by the user, making it practically impossible for an attacker to interrupt the authentication process. Server generates the server session and sends the server session's public key to the client. The client now generates the client session by decrypting the server session's public key. The client session's public key is now sent to the server, where the server processes it with the salt. Now, as the server has both the sessions, it tests for the validation of both the session keys. If validation is complete the user gets authenticated. Fig 5, denotes the general working flow of the authentication phase of our proposed system.

VIII. NORMAL AUTHENTICATION VS PROPOSED SYSTEM

To avoid retrieving the passwords in hashed or plaintext form, or any useful information by sniffing the network is

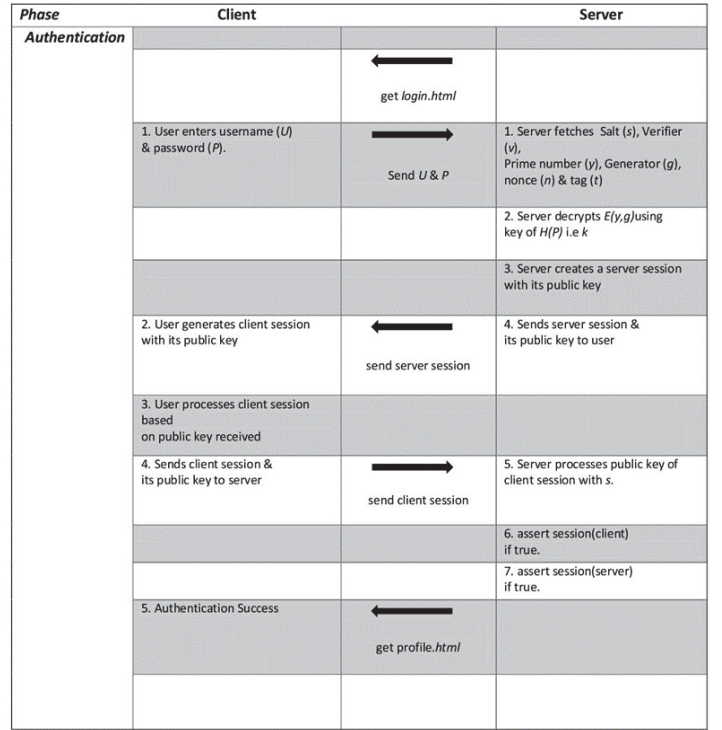


Fig. 5. General structure of the Authentication Phase.

important from a security perspective. It must be kept in mind that even if some amount of information gets exposed to the attacker, it must not be useful from the attacker's perspective, so that the attacker is not able to compromise the system.

A. Advantages of proposed system

- No hardware required like biometrics & token generator.
- Authentication is done without the password being sent over.
- The password stored on the server is encrypted.
- Using standard public key cryptographic algorithms like AES, provides a better secured authentication system.

IX. RESULTS & DISCUSSIONS

A. Security Analysis

In our proposed system, the secured approach to authenticate a user, is created using the concept of zero knowledge of proof. During authentication process, no information is disclosed. Thus, the attacker fails to retrieve the sensitive data as messages are sent from client to the server. Hence, our proposed system acts as a countermeasure for various common attacks that are discussed below:

1) *Replay Attack*: Replay attack is when, the authorized messages are captured by an attacker from the authenticated user and transmits it to act as an authorized user. As in our proposed system the random values are selected as verifiers for every new execution, replay attack is not possible. If an attacker passes the old captured message, the authentication will be failed.

2) *Phishing Attack*: Phishing is one of the most common attacks. In this attack, the attackers tempt and misguide the users to insert their confidential information on a fake website, which looks and feels almost indistinguishable to the legitimate one. In our proposed system, Before the data is sent to the server, the client needs to verify the challenge with . If the client fails to prove the challenge, the client will not be able to send the data to the server.

3) *Man-in-the-Middle Attack*: A sceario where the malicious user impersonates either of one party in a two party communication and modifies or relays the messages sent. In our proposed system, the attacker cannot gain any sensitive information, as no information is disclosed in zero knowledge proof during the authentication process.

4) *Sniffing*: The attacker sniffs passively through the communication channel, in order to gain the sensitive information of the user. As our system is based on zero knowledge proof, it makes the system to be zero knowledge. Despite the verifier having access to all the sensitive information, the attacker cannot read it.

5) *Password Guessing Attack*: Any traditional password based authentication system is vulnerable to the password guessing attack. Our proposed system acts as a countermeasure to this attack. The password is not sent over the channel in the system. The attacker cannot get any sensitive information from the two party communication in client and the server.

X. COMPARISON ANALYSIS

Table 2, gives a representation of comparative analysis on the time taken by the traditional authentication method and our proposed system, to register and authenticate a set of various users.

User ID	Registration time (in sec)	Normal Authentication Time (in sec)	Our ZKP Authentication Time (in sec)
User 1	8.322	5.062	4.753
User 2	8.462	4.321	3.435
User 3	8.314	5.422	3.879
User 4	7.343	5.834	3.538
User 5	9.824	5.145	3.146

TABLE II: Comparison analysis on time required for registration & authentication.

We have limited the scope of this paper in order to find a better futuristic authentication method that can be used in web applications. We have conducted a small-scale analysis based on different parameters, and taking into consideration the usability, security & deployability for our proposed system. Fig. 6, represents the comparative analysis done by using the USD Evaluation Framework, on the traditional authentication system and our proposed ZKP based authentication system.

XI. CONCLUSION

A thorough survey was done on various aspects of zero knowledge proof. ZKP can be implemented in a number of

Sr. No.	USD Evaluation Framework	Parameter	Our Proposed System	Traditional authentication system
1	Usability	Easy to learn/use	✓	–
		Easy loss recovery	✓	✓
		Efficiency	–	✓
		Memory Efficient	–	✓
2	Security	Resistant to leaks	✓	–
		Resistant to Sniffing	✓	–
		Resilient to dictionary attacks	✓	–
		No third party involved	✓	✓
		No explicit consent	✓	✓
3	Deployability	Browser Compatibility	✓	✓
		Server-compatible	✓	✓
		Accessibility	✓	✓
		Non proprietary	✓	✓

Fig. 6. Comparative analysis done by using the USD Evaluation Framework, on the traditional authentication system and our proposed ZKP based authentication system.

ways and applications. We used ZKP to build an authentication system. SRP and AES are used to build the ZKP based authentication system. AES being used on top of SRP. The proposed system overcomes a number of drawbacks of traditional and more commonly used authentication systems such as simple username, plaintext password based system, authentication system among others. Apart from the fact that our proposed system provides a higher degree of security & preserving the user's privacy at a higher level, various reasons like it's simple design and efficiency in regards of usability makes our proposed system worth to be integrated in web applications.

XII. FUTURE WORK

We have proposed an authentication system based on ZKP. Additional layers of security can be provided by implementing this system as a multi-factor authentication system. Our proposed system ensures that password does not travel from client to server, so this system can also be used where anonymity is to be ensured, such as for IoT devices authentication.

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